Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________ 11. ________

Transformations

Example (Describe the transformation & give the new vertex):
$g(x) = (x+4)^2 - 1$: inside $+4$ shifts **left 4**, outside -1 shifts **down 1**, so the vertex is $(-4, -1)$.

Directions: Describe each transformation from the parent x^2 and give the new vertex.

1. $y = (x-2)^2$

2. $y = x^2 + 3$

3. $y = (x+5)^2 - 2$

4. $y = -(x-1)^2$

5. $y = (x-3)^2 + 4$

6. $y = x^2 - 6$

7. $y = (x+2)^2$

8. $y = -(x+4)^2 + 1$

9. $y = (x-1)^2 - 5$

10. $y = -x^2 + 2$

Challenge (same sheet):

11. $y = -(x+2)^2 + 5$ — describe every transformation, give the vertex, and state whether it opens up or down.

Answer Key — fully worked

Procedure: Read the inside $(x-h)$: shift opposite the sign (left/right); Read the outside $+k$: shift up/down; A leading $-$ reflects over the x-axis; Vertex lands at (h,k).

1. Right 2; vertex $(2,0)$

2. Up 3; vertex $(0,3)$

3. Left 5, down 2; vertex $(-5,-2)$

4. Right 1, reflect over x-axis; vertex $(1,0)$, opens down

5. Right 3, up 4; vertex $(3,4)$

6. Down 6; vertex $(0,-6)$

7. Left 2; vertex $(-2,0)$

8. Left 4, up 1, reflect over x-axis; vertex $(-4,1)$, opens down

9. Right 1, down 5; vertex $(1,-5)$

10. Up 2, reflect over x-axis; vertex $(0,2)$, opens down

11. Left 2, up 5, reflect over x-axis; vertex $(-2,5)$, opens down